

IT-SIMPLERA INSTITUTE



Cyber Security Analyst Internship Program

WEEK 03 ASSIGNMENT

Computer Networking Fundamentals
& Network Configuration

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1. Introduction

This report is submitted as part of Week 3 of the Cyber Security Analyst Internship Program at ITSIMPLERA Institute. The objective of this assignment is to build a solid foundation in computer networking and core cybersecurity concepts through both theoretical research and hands-on practice using Kali Linux.

The report covers common network protocols and their default port numbers, a comparative analysis of HTTP versus HTTPS across ten live websites, hands-on network reconnaissance using ping and nmap, a detailed breakdown of what happens when a URL is entered into a browser, a professional networking cheat sheet, and web/domain enumeration using whatweb and whois. Together, these tasks develop practical command-line skills, analytical thinking, and the reporting discipline expected of a Cyber Security Analyst.

2. Task 1 — Protocols and Port Numbers

The table below lists 15 common network protocols along with their default port number, transport layer (TCP/UDP), purpose, and whether the protocol is considered secure or insecure by default.

Protocol	Port	TCP/UDP	Purpose	Secure or Insecure
FTP	21	TCP	Transfers files between client and server	Insecure — credentials & data sent in plain text
SSH	22	TCP	Encrypted remote login and command execution	Secure
Telnet	23	TCP	Unencrypted remote login to another host	Insecure — plain text credentials
SMTP	25	TCP	Sends email between mail servers	Insecure (unless STARTTLS/TLS is used)
DNS	53	TCP/UDP	Resolves domain names to IP addresses	Insecure by default (unless DNSSEC/DoH/DoT used)
DHCP	67/68	UDP	Automatically assigns IP addresses to devices	Insecure — no built-in authentication
HTTP	80	TCP	Transfers web pages between browser and server	Insecure — unencrypted
POP3	110	TCP	Downloads email from a mail server to a client	Insecure (unless POP3S/TLS used)
NTP	123	UDP	Synchronizes system clocks over a network	Insecure — can be abused for DDoS amplification
IMAP	143	TCP	Retrieves and manages email on a mail server	Insecure (unless IMAPS/TLS used)
SNMP	161	UDP	Monitors and manages network devices	Insecure in v1/v2c (plain text community strings)
LDAP	389	TCP	Accesses and maintains directory services	Insecure (unless LDAPS/TLS used)
HTTPS	443	TCP	Transfers web pages securely using TLS/SSL encryption	Secure

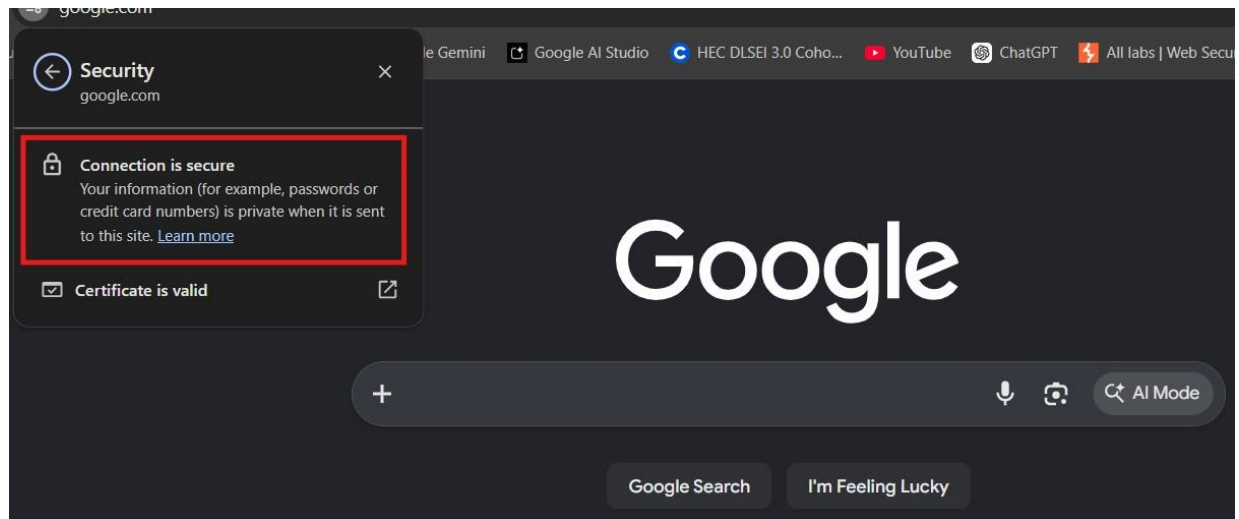
SMB	445	TCP	Shares files and printers on Windows networks	Insecure — historically targeted by ransomware (e.g., WannaCry)
Protocol	Port	TCP/UDP	Purpose	Secure or Insecure
RDP	3389	TCP	Provides graphical remote desktop access to Windows systems	Insecure if exposed to internet without MFA/VPN

3. Task 2 — HTTP vs HTTPS Analysis

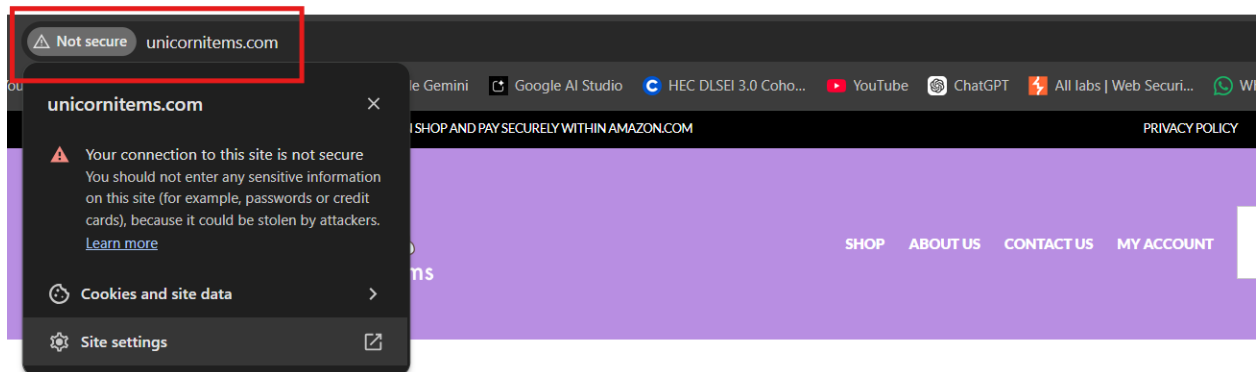
Ten major websites were analyzed to check whether they use HTTP or HTTPS, whether they automatically redirect from HTTP to HTTPS, and whether the connection is secure.

Website	HTTP or HTTPS	Auto-Redirects to HTTPS?	Is it Secure? (Explanation)
google.com	HTTPS	Yes	Secure — valid TLS certificate, enforces HSTS so browsers connect via HTTPS automatically.
github.com	HTTPS	Yes	Secure — strong TLS configuration, HSTS enabled, widely trusted CA-issued certificate.
wikipedia.org	HTTPS	Yes	Secure — HTTPS enforced site-wide since 2015, protects reader privacy and article integrity.
microsoft.com	HTTPS	Yes	Secure — valid certificate, redirects any HTTP request to HTTPS immediately.
amazon.com	HTTPS	Yes	Secure — TLS encryption protects payment and account data during checkout.
apple.com	HTTPS	Yes	Secure — enforces HTTPS across the whole domain with a modern TLS configuration.
linkedin.com	HTTPS	Yes	Secure — valid certificate; professional/personal data protected in transit.
x.com (Twitter)	HTTPS	Yes	Secure — HTTPS enforced, protects login sessions and DMs from interception.
yahoo.com	HTTPS	Yes	Secure — HTTPS with redirect from HTTP, valid CA-signed certificate.
Unicornitems.com	HTTP	No	InSecure

Google.com(HTTPS):



Unicornitems.com(HTTP)



Shop

CATEGORIES

No product categories exist.



4. Task 3 — Kali Linux Network Reconnaissance

4.1 Ping — Connectivity Check

Command used:

```
ping google.com
```

This sends 4 ICMP Echo Request packets to google.com and reports the response time and Time to Live (TTL) for each reply.

```
(kali㉿kali)-[~]
$ ping google.com
PING google.com (142.250.202.14) 56(84) bytes of data.
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=1 ttl=64 time=38.2 ms
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=2 ttl=64 time=43.1 ms
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=3 ttl=64 time=40.0 ms
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=4 ttl=64 time=42.6 ms
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=5 ttl=64 time=40.2 ms
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=6 ttl=64 time=41.7 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 38.230/40.963/43.106/1.679 ms
```

Metric	Value / Notes
Response Time (ms)	38.2ms
TTL Value	64
Interpretation	A low, stable response time indicates a healthy connection. The TTL value hints at the number of network hops / the remote OS since each router hop decrements TTL by 1.

4.2 Nmap — Service Discovery

Command used against scanme.nmap.org :

```
nmap -sV scanme.nmap.org
```

Explanation of the -sV :

The -sV flag tells Nmap to perform version detection: after finding an open port, Nmap probes it further to determine the specific service and version number running (e.g., 'Apache httpd 2.4.41' instead of just 'port 80 open'). This is important from a security-analyst perspective because a specific version can be checked against known CVEs (vulnerabilities) — a generic 'open port' tells you far less than a fingerprinted service and version.

```
(kali㉿kali)-[~]
$ nmap -sV -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-14 07:41 EDT
Warning: 45.33.32.156 giving up on port because retransmission cap hit (6).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.11s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
1720/tcp  filtered  h323q931
5060/tcp  filtered  sip
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.30 seconds
```

Finding	Details (fill in from your scan)
Open Ports Found	22/tcp , 25/tcp , 80/tcp , 1720/tcp , 5060/tcp, 9929/tcp , 31337/tcp
Services Running	22=ssh , 25=smtp , 80=http , 1720=h32q931 , 5060=sip , 9929=npingecho, 31337=tcpwrapped
Highest-Risk Open Port	22=OpenSSH 6.6.1p1 Ubuntu(SSh) , 80= Apache httpd 2.4.7(HTTP)

5. Task 4 — What Happens When You Type a URL and Press Enter?

When a user types a URL such as `https://www.example.com` into a browser and presses Enter, a sequence of well-defined steps occurs behind the scenes:

STEP 1: DNS Lookup

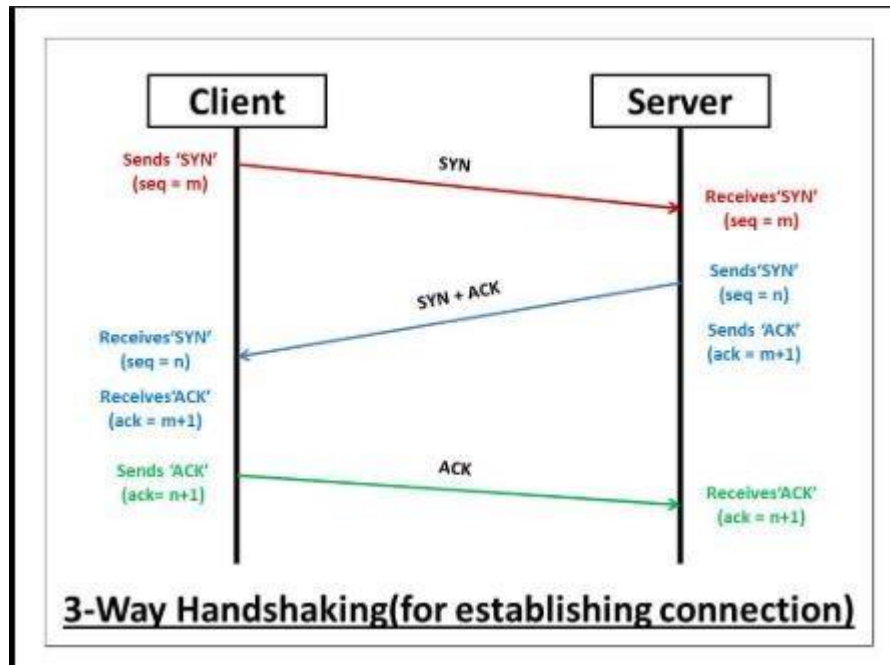
The browser first checks whether it already knows the IP address for the domain, checking (in order) its own cache, the OS cache, and the router/ISP resolver's cache. If not cached, a DNS query is sent to a recursive resolver, which in turn queries root servers, then Top-Level Domain (TLD) servers (e.g., .com), and finally the domain's authoritative name server to obtain the correct IP address.

STEP 2: IP Address Resolution

Once the authoritative name server responds, the browser (and OS) receives the resolved IP address (IPv4 or IPv6) and caches it for a period defined by the DNS record's TTL, so future requests to the same domain are faster.

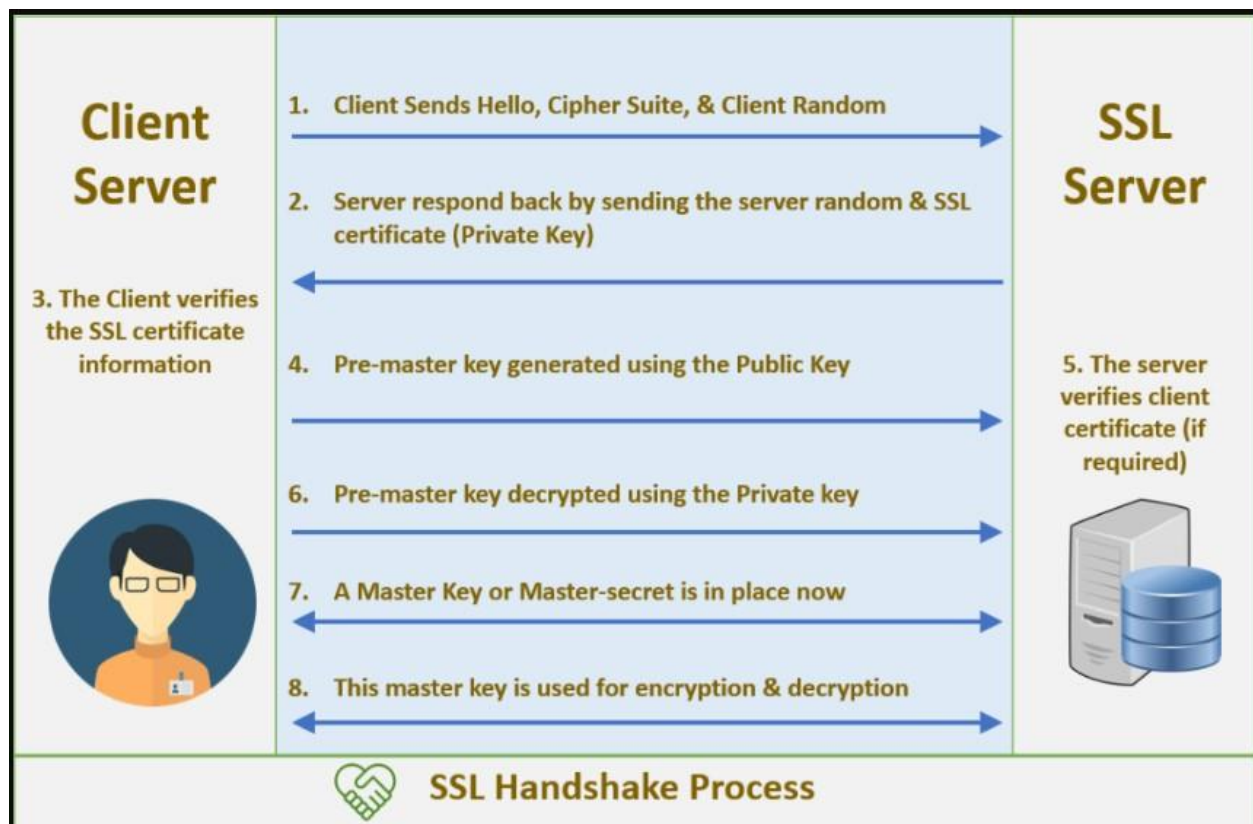
STEP 3: TCP Three-Way Handshake

Before any data is exchanged, the browser establishes a reliable TCP connection with the web server using a three-step process: the client sends a SYN (synchronize) packet, the server replies with a SYN-ACK (synchronize-acknowledge), and the client responds with an ACK (acknowledge). This confirms both sides can send and receive data, typically on port 443 for HTTPS or port 80 for HTTP.



STEP 4: TLS/HTTPS Handshake

For HTTPS sites, a TLS handshake follows the TCP handshake: the client and server agree on a TLS version and cipher suite, the server presents its SSL/TLS certificate (signed by a trusted Certificate Authority), the client verifies the certificate's validity and identity, and both sides derive a shared symmetric session key (via asymmetric key exchange, e.g., Diffie-Hellman) that will be used to encrypt all further communication.



STEP 5: HTTP Request

Once the secure (or plain) connection is established, the browser sends an HTTP request to the server — typically a GET request — including the requested resource path, headers (such as User-Agent, AcceptLanguage, and Cookies), and the HTTP method.

STEP 6: Server Response

The web server processes the request (often involving an application server, database queries, or API calls) and returns an HTTP response containing a status code (e.g., 200 OK, 404 Not Found, 301 Redirect), response headers, and the requested content — typically an HTML document plus references to CSS, JavaScript, and image files.

STEP 7: Browser Rendering the Webpage

The browser parses the HTML to build the DOM (Document Object Model), fetches and applies linked CSS to build the CSSOM, executes JavaScript, combines DOM and CSSOM into a render tree, and finally paints pixels to the screen — a process that may repeat (reflow/repaint) as additional resources load or scripts modify the page.

6. Task 5 — Networking Cheat Sheet

A quick-reference cheat sheet of 15 common network protocols, their ports, transport type, purpose, associated security risk, and a secure alternative where applicable.

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
FTP	21	TCP	File transfer	Credentials/data sent in clear text; sniffable	SFTP / FTPS
SSH	22	TCP	Secure remote administration	Weak passwords / brute force if exposed	Key-based auth + fail2ban
Telnet	23	TCP	Remote login (legacy)	Fully unencrypted — credentials easily captured	SSH
SMTP	25	TCP	Sending email	Open relays, spoofing, plaintext by default	SMTPS / STARTTLS
DNS	53	TCP/UDP	Domain name resolution	Cache poisoning, spoofing, amplification attacks	DNSSEC, DoH/DoT
DHCP	67/68	UDP	Automatic IP address assignment	Rogue DHCP servers, IP starvation attacks	DHCP snooping
HTTP	80	TCP	Web browsing	Data sent in plain text; MITM sniffing	HTTPS
POP3	110	TCP	Email retrieval	Plaintext credentials in transit	POP3S
NTP	123	UDP	Time synchronization	DDoS amplification via monlist	NTP with restricted queries
IMAP	143	TCP	Email retrieval & sync	Plaintext credentials in transit	IMAPS
SNMP	161	UDP	Network device monitoring	Default community strings ('public'/'private')	SNMPv3 with authentication
LDAP	389	TCP	Directory service access	Plaintext bind credentials	LDAPS
HTTPS	443	TCP	Secure web browsing	Misconfigured certs, weak TLS versions	Enforce TLS 1.2/1.3 only

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
SMB	445	TCP	Windows file/printer sharing	Ransomware propagation (e.g., WannaCry, EternalBlue)	SMB signing, patching, firewall block from WAN
RDP	3389	TCP	Remote desktop access	Brute force, exposed to internet, BlueKeep-type flaws	VPN + MFA, Network Level Authentication

7. Task 6 — Kali Linux Web Enumeration

The following tools were used to gather publicly available OSINT about google.com and github.com.

7.1 whatweb

```
whatweb google.com whatweb
unicornitems.com
```

```
(kali@kali)-[~]
$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.202.142], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[142.250.202.142], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID,__Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.151.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://www.google.com/ [200 OK] Cookies[AEC,NID,__Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.151.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]

(kali@kali)-[~]
$ whatweb unicornitems.com
http://unicornitems.com [403 Forbidden] Apache, Cookies[PHPSESSID], Country[UNITED STATES][US], DublinCore, Email[Artboard@2x.png,sprites@2x.png], HTML5, HTTPServer[Apache], IP[107.180.51.21], JQuery[1.12.4,3.3.1], MetaGenerator[Powered by Visual Composer - drag and drop page builder for WordPress,WooCommerce 3.5.2,WordPress 5.0.18], PHP[7.3.33], PoweredBy[Visual], Script[application/ld+json,text/javascript], Title[Unicorn Items], UncommonHeaders[link,upgrade], WordPress[5.0.18], X-Powered-By[PHP/7.3.33]
https://unicornitems.com [403 Forbidden] Apache, Bootstrap, Country[UNITED STATES][US], HTML5, HTTPServer[Apache], IP[107.180.51.21], JQuery[1.12.4], MetaGenerator[WordPress 4.9.22], PHP[7.1.33], Script[text/javascript], Title[Michael Nance &#8211; This website is under construction. Please check back soon!], UncommonHeaders[link,upgrade], WordPress[4.9.22], X-Powered-By[PHP/7.1.33]
```

7.2 whois

```
whois google.com whois
github.com
```

Field	google.com	github.com
-------	------------	------------

Domain Registrar	MarkMonitor Inc.	GoDaddy.com, LLC
Registration Date	1997-09-15T04:00:00Z	January 03, 2017 (2017-01-03)
Name Servers	NS1.GOOGLE.COM	NS35.DOMAINCONTROL.COMNS36.DOMAINCONTROL.COM
Interesting Finding		clientDeleteProhibited,clientRenewProhibited,clientTransferProhibited,clientUpdateProhibited

```
(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T05:19:04Z <<<

(kali@kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-18T05:19:04Z <<<
```

9. Challenges Faced

- [e.g., Initial confusion differentiating TCP vs UDP defaults for certain protocols — resolved by cross-checking IANA's official port registry.]
- [e.g., Local IP nmap scan initially returned no open ports because the local firewall was blocking probes — resolved by checking UFW/iptables rules.]
- [Add/replace with the actual issues you personally ran into while completing this assignment.]

Note: Replace the placeholders above with the real obstacles you encountered — evaluators specifically look for genuine, first-hand troubleshooting notes here.

10. Conclusion

This assignment provided practical, hands-on exposure to core networking and cybersecurity concepts — from understanding protocol-level port assignments and the security trade-offs between HTTP and HTTPS, to actively using Kali Linux tools (ping, nmap, whatweb, whois) for reconnaissance and enumeration. Mapping out the full lifecycle of a web request, from DNS resolution through TLS negotiation to page rendering, reinforced how many layers of the network stack cooperate for something as simple as loading a webpage. These foundational skills — command-line fluency, structured documentation, and a security-first mindset when evaluating protocols and open ports — directly support the daily responsibilities of a Cyber Security Analyst.

11. References

- IANA Service Name and Transport Protocol Port Number Registry — iana.org/assignments/service-names-port-numbers
- Nmap Official Documentation — nmap.org/book/man.html
- Nmap Authorized Test Target — scanme.nmap.org
- Mozilla Developer Network — 'What happens when you navigate to a URL' — developer.mozilla.org
- Cloudflare Learning Center — HTTP/HTTPS, TLS Handshake, DNS — cloudflare.com/learning
- WhatWeb Project — github.com/urbanadventurer/WhatWeb
- ICANN WHOIS — whois.icann.org